

Federated FHIR terminology alignment for diabetes reports.

Judge-facing wording

FHIR IPS is the interoperable artifact; PDFs are human-readable renderings.

The Problem

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Local clinical phrases are ambiguous across countries: T2DM, sugar disease, raised BP, Typ-2-Diabetes. The problem is semantic meaning, not only file format.

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Why HL7

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FHIR IPS gives the patient-summary exchange artifact. FHIR terminology resources make mappings auditable through CodeSystem, ValueSet, ConceptMap, translate, lookup, and validate-code.

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Solution Flow

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Local report -> clinical facts -> registry lookup -> federated linker on misses -> FHIR terminology validation -> FHIR IPS Bundle -> target-country readiness and PDF rendering.

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Federated Learning Role

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Each site trains locally. The coordinator receives model tensors and sample counts only, not raw reports, aliases, labels, identifiers, or patient-level FHIR Bundles.

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Evidence

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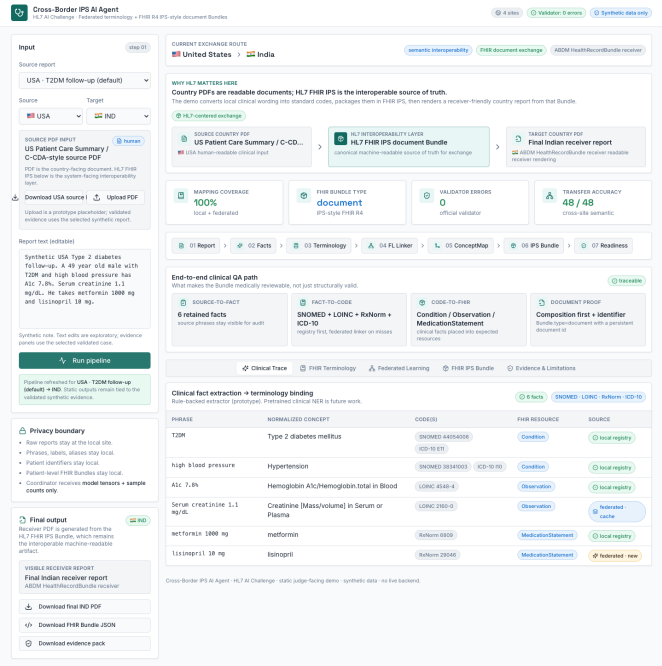
20 synthetic reports, 4 sites, 768 training examples, 48 cross-site transfer examples, 192 globally unseen examples, 48/48 transfer correct, 0 validator errors.

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Judge can test USA->India, India->USA, Australia->Europe, and Europe->USA.
Each route exports source PDF, final PDF, FHIR Bundle JSON, and evidence pack JSON.

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Dashboard screenshot from automated click-test.

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Honest Limits

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Synthetic data only. Rule-backed extraction. Simulated terminology operations. Readiness checks only. FedAvg gives data locality only, not formal privacy.

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Final Claim

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Federated learning resolves local terminology without centralizing raw reports. FHIR terminology artifacts make mappings auditable. FHIR IPS packages the result for exchange.

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